

EDUCATION

National Taiwan University (NTU)

M.S. in Electrical Engineering, GPA: 4.16/4.30

Sep 2024 – Aug 2026

Taipei, Taiwan

National Yang Ming Chiao Tung University (NYCU)

B.S. in Computer Science, GPA: 4.28/4.30, Rank: 2/201 (top 1%)

Sep 2020 – Jun 2024

Hsinchu, Taiwan

PUBLICATIONS

Y.-C. Huang, P.-C. Hsiu, R. Fang, and M.-S. Chen, “KV Admission: Learning What to Write for Efficient Long-Context LLM Inference,” *EMNLP*, 2026 (top NLP conference)

C.-Y. Liu, H. R. Mendis, **Y.-C. Huang**, Y.-J. Chen, and P.-C. Hsiu, “Splitting Bottlenecks: Memory-Aware Neural Architecture Search for Multi-Branch TinyML,” *IEEE TCAD*, 2026 (top EDA journal)

Y.-C. Huang, C.-P. Yang, and T. T. Yeh, “AQB8: Energy-Efficient Ray Tracing Accelerator through Multi-Level Quantization,” *ISCA*, 2025 (top computer architecture conference)

EXPERIENCE

Graduate Researcher

NTU & Academia Sinica (Advised by Prof. Ming-Syan Chen and Prof. Pi-Cheng Hsiu)

Sep 2024 – Aug 2026

Taipei, Taiwan

- Proposed KV Admission, a learned write gate that filters redundant tokens before cache entry, and built a dual-cache paged manager with sparse attention kernels for 1.6-4.2x faster long-context LLM inference (EMNLP 2026).
- Built the TensorFlow Lite Micro and STM32Cube.AI deployment toolchain for DupNAS, a memory-aware neural architecture search (NAS) framework for multi-branch TinyML on STM microcontrollers (IEEE TCAD 2026).

Undergraduate Researcher

National Yang Ming Chiao Tung University (Advised by Prof. Tsung Tai Yeh)

Jul 2022 – Jun 2024

Hsinchu, Taiwan

- Designed AQB8, a ray tracing (RT) accelerator that traverses BVH trees directly in low-bit integers via multi-level quantization, achieving 49% lower energy and 27% smaller area over modern GPU RT accelerators (ISCA 2025).

Teaching Assistant, Digital Circuit Design

National Yang Ming Chiao Tung University (English-medium course)

Feb 2023 – Jun 2023

Hsinchu, Taiwan

- Designed Verilog lab assignments and ran oral demo sessions, questioning students on their implementations to verify understanding; graded exams, homework, and lab reports while supporting students in office hours.

HONORS & AWARDS

Outstanding Students Conference Travel Grant (for presenting at ISCA 2025)

FAOS, 2025

Phi Tau Phi Scholastic Honor Society Member (for top 1% in CS department)

NYCU, 2024

Yin Zhi Tong Electronics Laboratory Scholarship (for excellence in electronics project)

NYCU, 2024

Best Project Award, Competition of Computer Science and Engineering Projects

NYCU, 2024

Academic Excellence Award (for top 5% in CS department, awarded in 7 semesters)

NYCU, 2020 – 2024

SKILLS

Programming: C/C++, Python, Verilog/SystemVerilog, CUDA, Triton, OpenMP, MPI, SystemC, R, Bash, Tcl

ML Systems: PyTorch, Hugging Face Transformers/PEFT, FlashAttention, TensorFlow Lite Micro, STM32Cube.AI

Hardware Design: Vivado, Vitis, PetaLinux, Zynq MPSoC, Catapult HLS, Design Compiler, QuestaSim, VCS/Verdi

Languages: Chinese (native), English (TOEIC 920, GEPT High-Intermediate)

SELECTED PROJECTS

RISC CPU Design, from RTL to GDSII: Independently built a 16-bit MIPS-like CPU with AXI4 memory interfaces, then took it through synthesis, floorplanning, automatic place & route, post-layout simulation, STA, LVS, and DRC.

Channel, Global, and Detailed Routers for VLSI: Built a greedy channel router, a global router using negotiation-based rip-up and reroute on ISPD 2008 benchmarks, and a detailed router that encodes routability as MaxSAT.

Linux Kernel Module and Page-Table Manipulation: Hooked the syscall table to hide the module and filter syscalls per process; added an ARM64 syscall and a `userfaultfd` handler that install page-table entries on demand.

Multimodal VLM for Driving Comprehension: Led the team to 1st of 12 teams with a LoRA-tuned LLaVA fusing segmentation, depth, and ViT cues by cross-attention, plus a self-critiquing loop mining new QA pairs from its mistakes.

Additional projects, with code and details, at lashhw.github.io/projects